TECHNICAL CAPABILITY GUIDE

SpeedCloud

Capability catalog explained

A practical guide to selecting compute, networking, data, continuity, migration, security, and identity capabilities from the SpeedCloud catalog.



Source visual reproduced from SpeedCloud_Capability_Catalog_July2026.pdf. Service names, limits, availability, and commercial terms should be confirmed for the target design.

01 | CATALOG AT A GLANCE

A service map for solution design

The catalog is a capability-level map. It groups services by the problem they solve and shows comparable offerings from major cloud providers. Use it to form an initial architecture, then validate exact service limits, regions, support boundaries, security controls, and pricing before implementation.

Family	Capabilities listed in the catalog	Typical design question
Compute	General-purpose virtual machines; bare-metal servers.	Do we need flexible virtual capacity, or dedicated hardware and isolation?
Networking	Virtual networks, private links, remote and site-to-site VPN, CDN, load balancing, autoscaling, SIP trunk.	How will users, branches, services, and private dependencies connect securely?
Cloud-native and data	Managed Kubernetes; relational and non-relational database services.	Should the platform operate the control plane and database lifecycle for us?
Storage and resilience	NVMe block, S3-style object, private unified storage, managed backup, disaster recovery.	What are the access pattern, recovery objective, retention, and portability needs?
Migration and protection	VM/workload migration; firewall, IDS/IPS, WAF, DDoS, endpoint security, KMS/HSM, IAM.	How do we move workloads and apply layered controls without leaving responsibility gaps?

How to read the service labels

- IaaS usually means the customer still owns more of the operating system, application, and data configuration.
- PaaS generally moves more platform operations to the provider, while the customer remains responsible for application behavior and data.
- SaaS places the service closest to the provider-managed application layer; identity, data use, and configuration still need governance.
- The AWS, Google Cloud, and Azure entries are capability analogies for orientation, not proof of identical features or service parity.

A good design starts with the workload's operating model, not with a product name.

02 | COMPUTE SERVICES

Choose the execution model before the instance size

The catalog describes general-purpose instances and bare-metal servers. The practical difference is not only speed; it is the level of abstraction, isolation, operational effort, and tuning control that the workload requires.

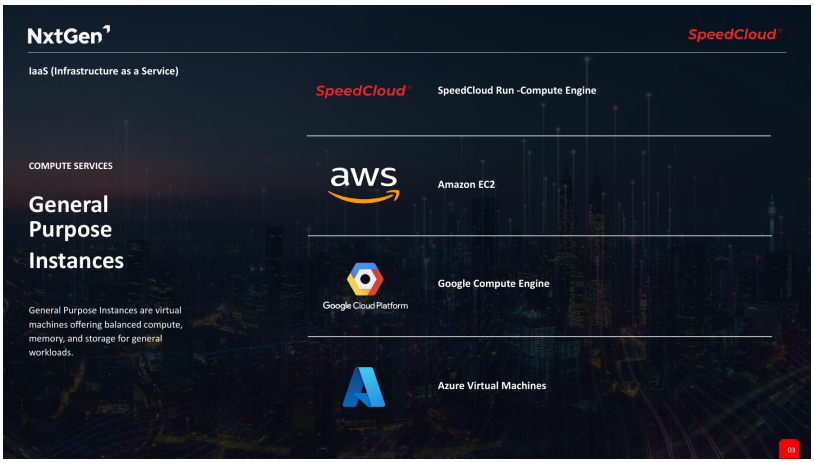


Figure 1. General-purpose compute positioning from the source catalog.

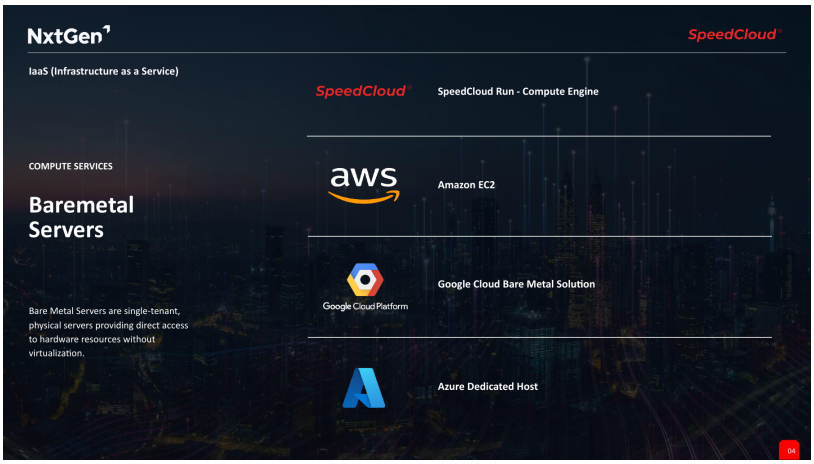


Figure 2. Bare-metal compute positioning from the source catalog.

Option	Use it when	Example workload
General-purpose VM	You need balanced compute, memory, and storage with fast provisioning and flexible scaling.	Web/API tiers, business applications, development environments, and most standard VM migrations.
Bare metal	You need single-tenant physical resources, strong isolation, direct hardware access, or predictable performance without virtualization.	Licensed software, high-I/O databases, specialized appliances, and workloads with strict tenancy or tuning requirements.

Technical checks

- Confirm vCPU model, CPU allocation, memory ratio, local or attached storage, network bandwidth, and burst behavior.
- For bare metal, confirm provisioning time, hardware replacement process, firmware responsibility, and supported hypervisors or operating systems.

03 | NETWORKING

Build connectivity in layers

The catalog separates private cloud networking, dedicated private links, remote access, site-to-site connectivity, delivery, traffic distribution, and automated capacity. These are complementary controls rather than interchangeable products.

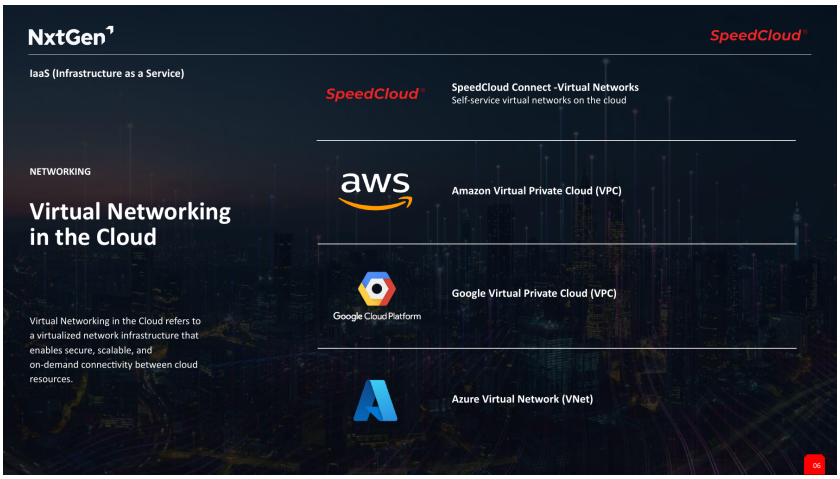


Figure 3. Virtual networking positioning from the source catalog.

Capability	Primary purpose	Example
Virtual network	Create isolated address space, subnets, routes, and security boundaries for cloud resources.	Separate web, application, database, and management subnets.
Private link	Connect services or sites over dedicated private paths without exposing traffic to the public internet.	Private interconnect to a data center, partner, or managed service.
Remote access VPN	Give individual users encrypted access to virtual workloads with authentication and policy controls.	Administrator access to a private management subnet.
Site-to-site VPN	Extend an on-premises or branch network to cloud networks over an encrypted tunnel.	Branch office access to a cloud-hosted ERP application.

04 | NETWORKING FOR APPLICATION BEHAVIOR

Deliver, distribute, and scale traffic

Once connectivity exists, applications need a controlled way to receive traffic, serve users near their location, and add or remove capacity as demand changes.

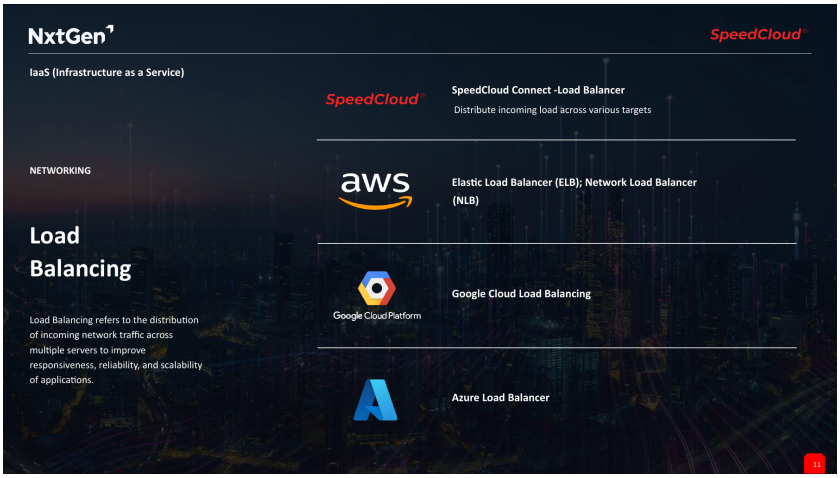


Figure 4. Load-balancing positioning from the source catalog.

Capability	What it changes	Test or metric
CDN	Caches and serves content closer to users, reducing origin distance and improving delivery latency.	Cache hit ratio, time to first byte, origin load, and regional latency.
Load balancer	Distributes requests across healthy targets and can provide a stable service endpoint.	Latency, error rate, health-check behavior, failover, and connection draining.
Autoscaling	Adjusts capacity based on demand or policy to balance performance and cost.	Scale-out time, scale-in safety, saturation, queue depth, and cost under peak load.
SIP trunk	Connects a private branch exchange to public telephony using SIP.	Call quality, redundancy, codec support, failover, and number portability.

Example: highly available web service

- Place application instances in separate failure domains behind a load balancer.
- Use autoscaling on a measured signal such as request rate, queue depth, or CPU saturation.
- Use a CDN for cacheable static content, while keeping sensitive or dynamic paths at the origin.
- Test health-check failure, target drain, zone loss, and sudden demand before production launch.

05 | CLOUD-NATIVE AND DATABASE SERVICES

Use managed control planes where operations are not the differentiator

The catalog includes managed Kubernetes plus relational and non-relational database services. These choices affect application design, operational ownership, data modeling, and recovery procedures.

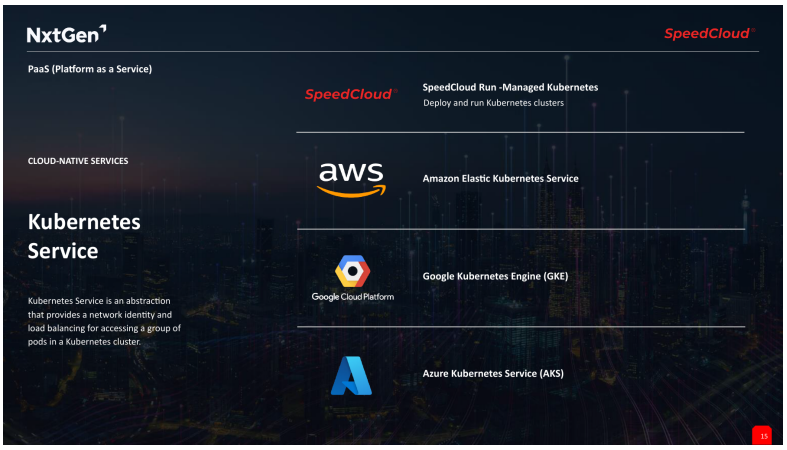


Figure 5. Managed Kubernetes positioning from the source catalog.

Service	Best fit	Important design questions
Managed Kubernetes	Containerized workloads needing orchestration, service discovery, rolling deployment, and scheduling.	Who manages nodes, upgrades, ingress, observability, policy, secrets, and persistent volumes?
Relational DBMS	Structured data, transactions, joins, referential integrity, and SQL reporting.	Engine/version, HA, backups, replicas, connection limits, maintenance windows, and recovery time.
Non-relational DBMS	Flexible-schema, key-value, document, or graph access patterns that scale around a defined data model.	Consistency model, partition key, indexes, query limits, backup/export, and migration path.

Example selection rule

- Choose relational storage when correctness depends on transactions and relationships.
- Choose NoSQL when access patterns are known, scale is driven by key or document access, and flexible schema is valuable.
- Choose managed Kubernetes when the team benefits from the orchestration platform and can operate the application layer effectively; Kubernetes is not automatically simpler.

06 | STORAGE, CONTINUITY, AND MIGRATION

Match the data service to access pattern and recovery objective

The catalog distinguishes block, object, private unified storage, backup, disaster recovery, and workload migration. These solve different problems and should be designed together.

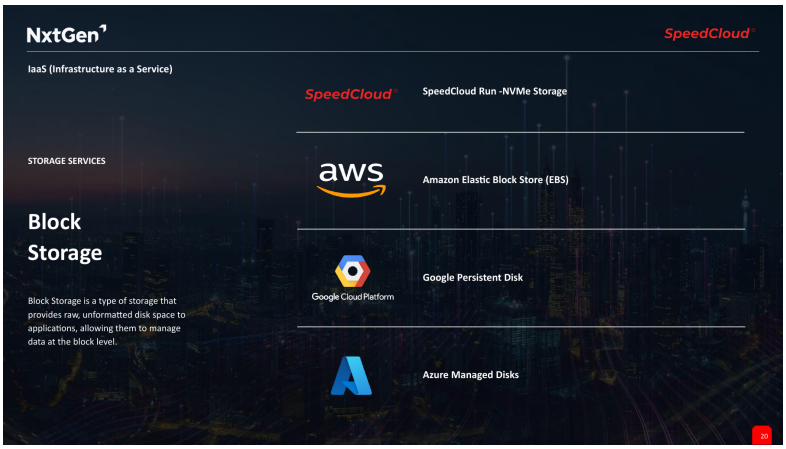


Figure 6. Block-storage positioning from the source catalog.

Capability	Use it for	Example
Block storage	Low-latency volumes mounted to compute and managed by the operating system or application.	Database data files, VM boot volumes, and transactional application storage.
Object storage	Highly durable objects accessed through APIs, with metadata and large-scale namespace.	Backups, media, data lakes, logs, artifacts, and static content.
Private unified storage	A private storage layer that brings block, file, and object access together for enterprise integration.	Shared enterprise data used by multiple protocols or connected environments.
Backup	Create recoverable copies for accidental deletion, corruption, or operational rollback.	Daily database backups with retention and restore verification.
Disaster recovery	Restore service after a larger outage with a defined recovery point and recovery time.	Replicated application stack with a tested failover runbook.
Migration	Move VMs and workloads between on-premises, cloud, or provider environments.	Assess, replicate, test, cut over, validate, and retain a rollback path.

Recovery checklist

- Write down RPO, RTO, retention, encryption, restore ownership, and evidence requirements.
- Test restore and failover, not only backup completion or replication status.
- Measure data transfer time, cutover downtime, application compatibility, and rollback readiness during migration.

07 | SECURITY AND IDENTITY

Apply controls at the network, application, endpoint, key, and identity layers

The catalog presents security as a stack. No single control replaces the others: a firewall filters traffic, a WAF protects web requests, IAM governs access, and key management protects cryptographic material.

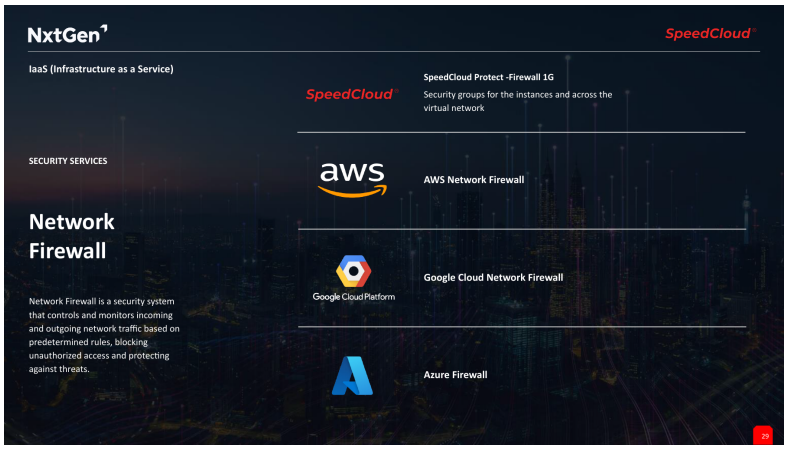


Figure 7. Network-firewall positioning from the source catalog.

Control	Primary role	Example use
Network firewall	Controls permitted network flows between zones, workloads, and interfaces.	Allow only required application ports from approved network segments.
IDS/IPS	Detects suspicious activity and can block or respond to known threats.	Alert on scans, exploit signatures, or policy violations and investigate the evidence.
WAF	Filters HTTP/S requests to protect web applications from common attacks and abuse.	Protect login, API, and public web routes with managed rules and tuning.
DDoS protection	Mitigates distributed traffic floods that exhaust bandwidth or service capacity.	Protect public endpoints and define escalation or scrubbing procedures.
Endpoint security / XDR	Monitors and responds to threats on user or server endpoints and correlates signals.	Protect administrator laptops and workload hosts with response workflows.
KMS/HSM	Creates, stores, controls, and audits encryption keys and sensitive cryptographic operations.	Separate key administration from data administration and test rotation or recovery.
IAM	Authenticates identities and authorizes least-privilege access to resources.	Use roles, MFA, federation, access reviews, and auditable service identities.

Example: production API

- Use a virtual network and firewall rules to limit reachable paths.
- Place a WAF and DDoS protection in front of public HTTP endpoints.
- Use IAM with MFA and short-lived roles for operators and deployment automation.
- Encrypt data and manage keys separately, with rotation, audit, and recovery procedures.

08 | PUTTING THE CATALOG TO WORK

Start with a workload, then assemble the smallest complete service set

The catalog is most useful when it becomes a design conversation. Select services by workload behavior, availability target, compliance requirements, operational ownership, and full cost rather than by feature count.

Scenario	Likely starting set	Success measures
Public web application	General-purpose VMs, virtual network, load balancer, CDN, autoscaling, WAF, IAM, backup.	p95 latency, error rate, scale response, availability, security findings, and cost per request.
Regulated enterprise system	Private network, site-to-site VPN or private link, managed database, KMS/HSM, IAM, backup, DR, audit evidence.	Residency, access audit, RPO/RTO, restore test, control ownership, and operational response.
High-I/O database	Bare metal or tuned compute, NVMe block storage, relational DBMS, private connectivity, backup, monitoring.	p95/p99 latency, IOPS, transaction rate, contention, backup impact, and cost per transaction.
Cloud migration	Migration service, target compute, network connectivity, storage mapping, IAM, security controls, backup, rollback plan.	Migration duration, downtime, data integrity, compatibility, rollback time, and runbook quality.
Container platform	Managed Kubernetes, virtual network, load balancer, private registry or object storage, IAM, observability, backup.	Deployment frequency, recovery, scaling, cluster operations effort, security posture, and application latency.

Proof-of-concept sequence

- Baseline the current workload: latency, throughput, storage I/O, utilization, availability, recovery, security, and cost.
- Map each required capability to a named SpeedCloud service and record what remains the customer's responsibility.
- Run normal, peak, failure, backup/restore, access-review, and migration or rollback scenarios.
- Review technical results, operator effort, complete cost, support model, portability, and exit path.

The catalog is a starting point for architecture; the signed service description and workload test are the acceptance criteria.

09 | DECISION CHECKLIST

Questions to close before production

Use this checklist to turn the catalog into an implementation decision that infrastructure, security, finance, application, and operations teams can all support.

Area	Questions to answer
Service fit	Which workload behavior requires this service, and what is the simplest alternative?
Responsibility	Who owns the operating system, patches, keys, identity, backups, monitoring, incident response, and upgrades?
Network	Which paths are public, private, encrypted, inspected, logged, and resilient to link failure?
Data	What are the access pattern, consistency model, retention, residency, backup, restore, and export requirements?
Security	Which controls are preventive, detective, or responsive, and where is the evidence stored?
Performance	Which metric proves the design meets the workload target under normal and peak conditions?
Continuity	Have restore, failover, rollback, and operator handover been tested with measured RPO and RTO?
Commercial	Does the total cost include data transfer, support, licenses, operations, migration, backup, and exit effort?

Technical conclusion

The SpeedCloud capability catalog covers the core building blocks of a cloud operating model: compute, connectivity, application delivery, data platforms, storage, resilience, migration, protection, and identity. The right design is the smallest complete set of services that meets the workload's technical and operational requirements with an explicit responsibility matrix.

All service names, feature limits, certifications, performance claims, regions, support targets, and prices should be confirmed against the current service schedule and signed commercial agreement.